

ViTs for Action Classification in Videos: An Approach to Risky Tackle Detection in American Football Practice Videos

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Performance Across Augmentation Runs

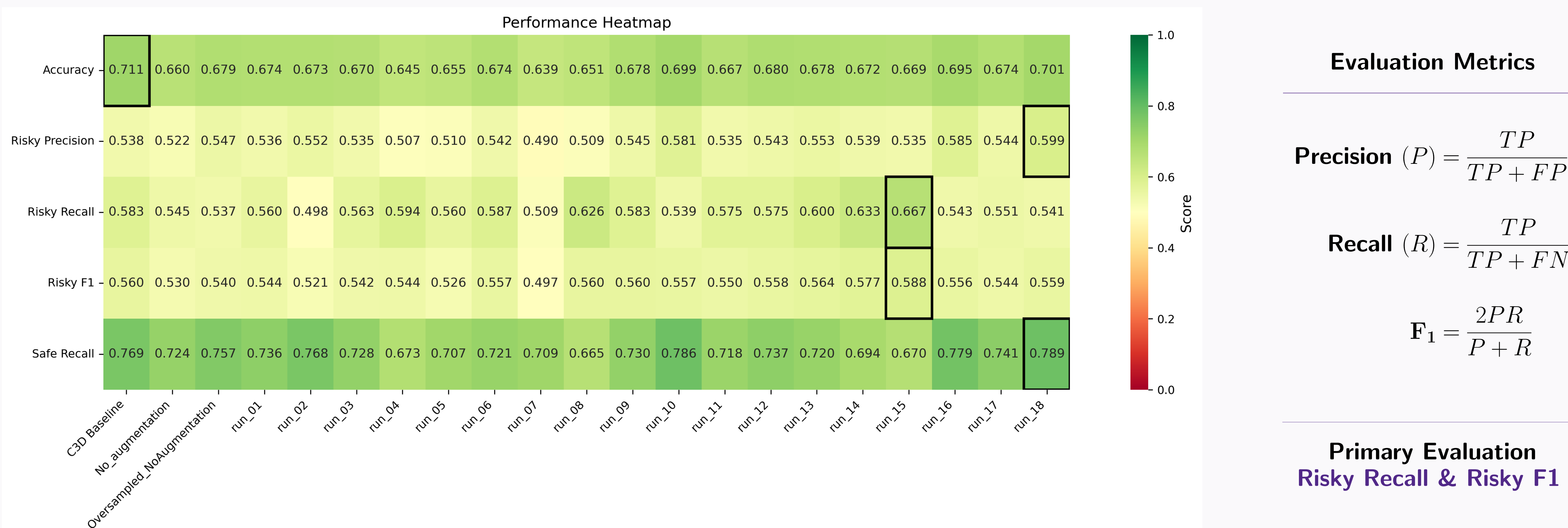


Figure 1. Performance across ViViT augmentation configurations. The prior C3D baseline is shown for historical comparison.

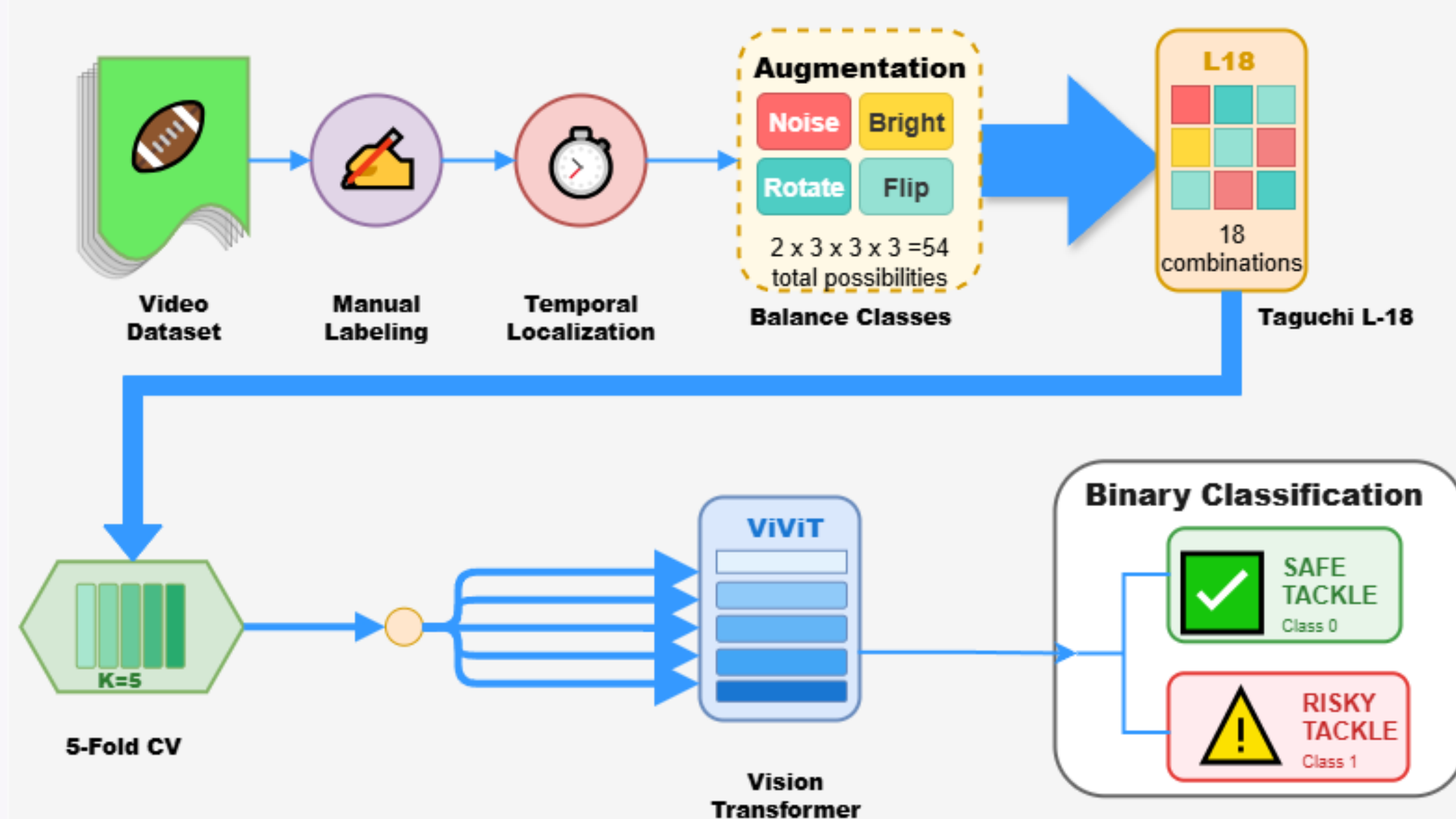
Problem Statement and Objective

Risky tackling techniques can increase the risk of head and cervical injury in American football. We investigate whether video Vision Transformers can improve the classification of risky tackles from practice videos using **SATT-3 strike-zone labels**.

- **Challenge:** Risky tackles are the minority class (259 risky vs. 474 safe), and practice videos contain substantial visual variability.
- **Objective:** Improve risky-tackle detection by combining temporal localization, targeted video augmentation, and ViViT-based classification.
- **Evaluation focus:** Prioritize **risky recall** and **risky F1** while monitoring the trade-off with overall classification performance.

Experimental Methodology

- Localized the **first point of contact** and extracted a fixed temporal window.
- Applied **Taguchi L18-guided augmentation** to the training videos.
- Fine-tuned a pretrained **ViViT** using 5-fold cross-validation.



Dataset and Augmentation Examples

Dataset Distribution

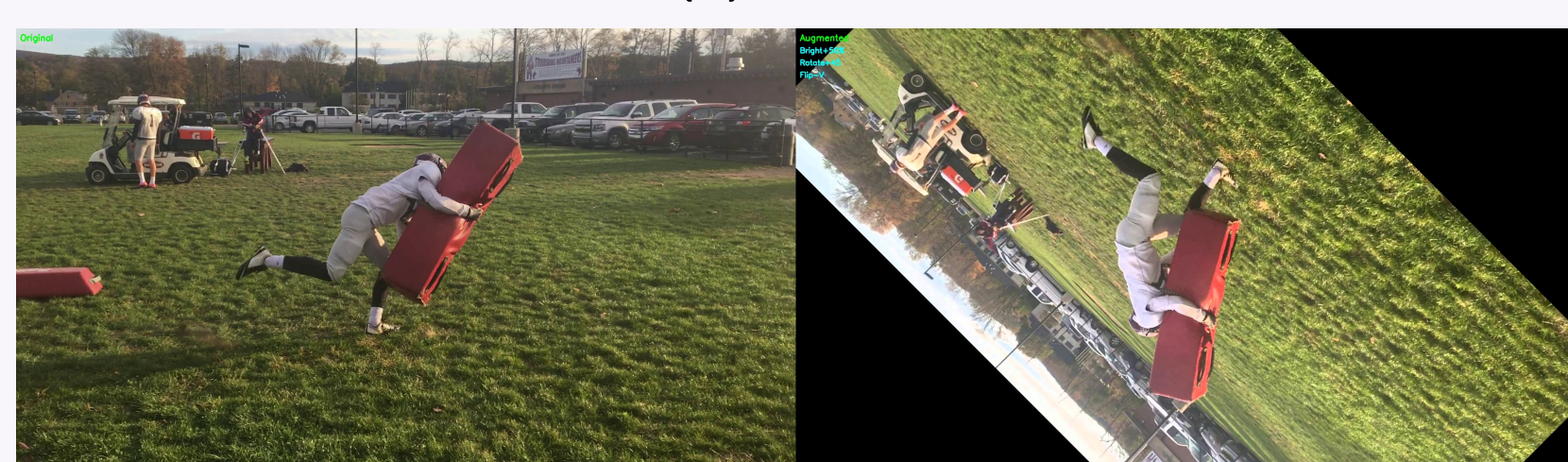
Class	Count	%
Safe	474	64.7
Risky	259	35.3
Total	733	100.0

Class imbalance: Risky tackles comprise only 35.3% of the dataset, motivating targeted augmentation of the minority class.

Original (left) vs. Augmented (Right) Frames



(a) Run 4



(b) Run 2

Representative original and augmented tackle frames under selected Taguchi augmentation configurations.

Taguchi Augmentation Experiment Design

Table 1: Taguchi L_{18} schedule for augmentation factors.

Run	Add noise (A)	Brightness (B)	Rotate (C)	Flip (D)
R1	None	Increase	Left	Horizontal
R2	None	Increase	Right	Vertical
R3	None	Increase	None	None
R4	None	Decrease	Left	Vertical
R5	None	Decrease	Right	None
R6	None	Decrease	None	Horizontal
R7	None	None	Left	None
R8	None	None	Right	Horizontal
R9	None	None	None	Vertical
R10	Add noise	Increase	Left	None
R11	Add noise	Increase	Right	Horizontal
R12	Add noise	Increase	None	Vertical
R13	Add noise	Decrease	Left	Horizontal
R14	Add noise	Decrease	Right	Vertical
R15	Add noise	Decrease	None	None
R16	Add noise	None	Left	Vertical
R17	Add noise	None	Right	None
R18	Add noise	None	None	Horizontal

Results:

Table 2: Performance metrics from the heatmap for **C3D Baseline**, **Run₁₅**, and **Run₁₈**

Metric	Prior C3D	Run 15	Run 18
Accuracy	0.711	0.669	0.701
Risky Precision	0.538	0.535	0.599
Risky Recall	0.583	0.667	0.541
Risky F1	0.560	0.588	0.559
Safe Recall	0.769	0.670	0.789

Note: The C3D baseline is from prior work on a smaller dataset; ViViT results are reported on the expanded 733-clip dataset and are not a same-test-set comparison.

- **Run 15** increased risky recall from **0.583 to 0.667** and achieved the highest risky F1 (**0.588**).
- This improvement trades off overall accuracy: **0.711** (C3D) vs. **0.669** (Run 15).
- **Run 18** recovered accuracy to **0.701** while achieving the highest risky precision (**0.599**) and safe recall (**0.789**).

Conclusion & Key Takeaways

- **Run 15** achieved the strongest risky-class performance: **0.667 risky recall** and **0.588 risky F1**.
- The best recall configuration used **added noise, decreased brightness, no rotation, and no flipping**.
- Augmentation choice changes the trade-off between **risky-tackle sensitivity** and **overall performance**.